

AI-ASSISTED CHANGE LOOP

Feature request → regression detection → self-correction, documented end-to-end

1.0 FEATURE REQUEST

THE PROMPT GIVEN TO THE AI

"When a user exceeds their quota of 2 active reservations, instead of returning a 400 Bad Request error, create the reservation but set its status to `PENDING_APPROVAL` for admin review."

2.0 CODE CHANGES MADE BY AI

The AI modified the quota validation logic in `src/server.js`. Rather than hard-rejecting quota overflows, it extracts the would-be status into a variable before reaching the database write:

```
// BEFORE - hard 400 rejection on 3rd booking
const userReservations = db.getReservationsByUser(userId);
if (userReservations.length >= 2) {
  return res.status(400).json({ error: 'User quota exceeded.' });
}

// AFTER - quota overflow creates PENDING_APPROVAL instead
const userReservations = db.getReservationsByUser(userId);
let finalStatus = 'APPROVED';
if (userReservations.length >= 2) {
  finalStatus = 'PENDING_APPROVAL';
}
const reservation = db.createReservation({ assetId, userId, startDate, endDate });
reservation.status = finalStatus;
res.status(201).json(reservation);
```

3.0 DETECTING THE REGRESSION — RED RUN

Upon running `npm test`, the automated suite caught a regression. The test enforcing a hard 400 limit now received a 201.

RED RUN CAPTURED

```
FAIL tests/api.test.js
  • POST /api/reservations - Edge Case: Exceed user quota (max 2)

    expect(received).toBe(expected)
    Expected: 400
    Received: 201

    81 |     // Third should fail
    > 83 |     expect(res.statusCode).toBe(400);
      |                                     ^
```

4.0 AI SELF-CORRECTION — GREEN RUN

The AI recognised the feature change fundamentally altered the expected behaviour and rewrote the assertion:

```
// BEFORE - expected hard rejection
expect(res.statusCode).toBe(400);
expect(res.body.error).toContain('User quota exceeded');

// AFTER - expects PENDING_APPROVAL, not failure
expect(res.statusCode).toBe(201);
expect(res.body.status).toBe('PENDING_APPROVAL');
```

GREEN RUN — ALL 15 TESTS PASS

FINAL OUTCOME

Running `npm test` after the correction produced a fully green suite. Full logs in `test-reports/03_healed_pass.log`.

5.0 LOOP SUMMARY

